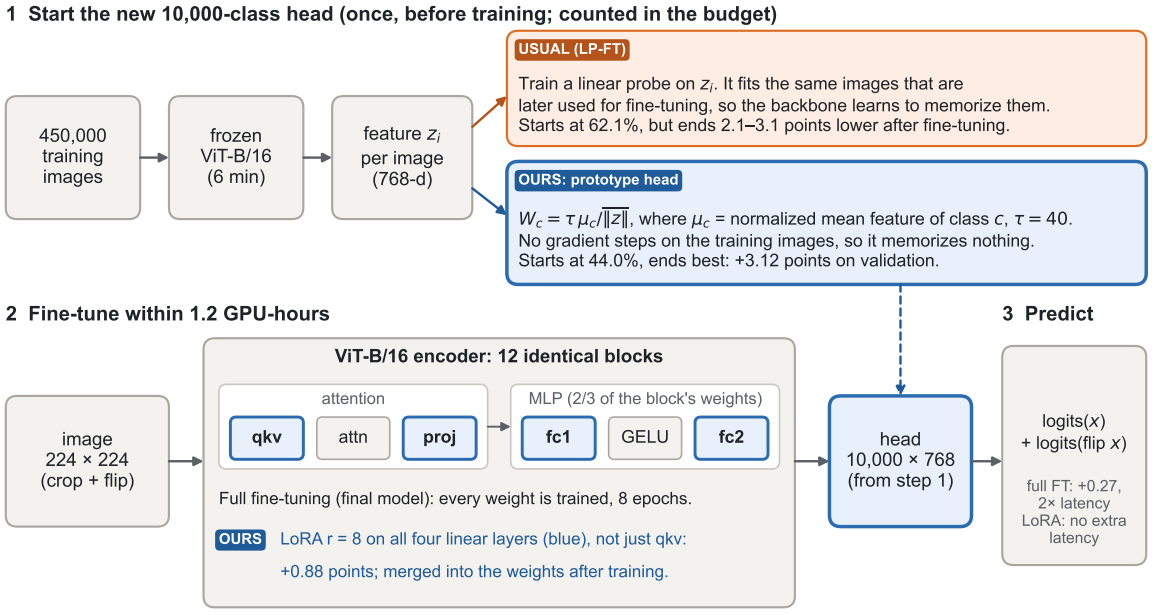
1 Start the new 10,000-class head (once, before training; counted in the budget)
450,000 training images
frozen ViT-B/16 (6 min)
feature z_i per image (768-d)
USUAL (LP-FT)
Train a linear probe on z_i. It fits the same images that are later used for fine-tuning, so the backbone learns to memorize them.
Starts at 62.1%, but ends 2.1–3.1 points lower after fine-tuning.
OURS: prototype head
$W_c = \tau \mu_c / \overline{\|z\|}$, where μ_c = normalized mean feature of class c, $\tau = 40$.
No gradient steps on the training images, so it memorizes nothing.
Starts at 44.0%, ends best: +3.12 points on validation.
2 Fine-tune within 1.2 GPU-hours
3 Predict
image 224 × 224 (crop + flip)
ViT-B/16 encoder: 12 identical blocks
attention
qkv
attn
proj
MLP (2/3 of the block's weights)
fc1
GELU
fc2
Full fine-tuning (final model): every weight is trained, 8 epochs.
OURS
LoRA r = 8 on all four linear layers (blue), not just qkv:
+0.88 points; merged into the weights after training.
head 10,000 × 768 (from step 1)
logits(x) + logits(flip x)
full FT: +0.27, 2× latency
LoRA: no extra latency